

# Shaoyuan Xie

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## EDUCATION

### University of California, Irvine

*Ph.D. in Computer Science, Advisor: Professor Qi Alfred Chen*

Irvine, CA

*Expected Jun 2028*

### Huazhong University of Science and Technology (HUST)

*B.Eng. in Automation, GPA: 3.97 (Top 1st / 212 students)*

Wuhan, CN

*Jun 2023*

## INTERNSHIPS

### Mercedes-Benz Research & Development North America

*AD Perception | Mentor: Thomas Monninger | Manager: Sihao Ding*

San Jose, CA

*Jun 2025 – Sep 2025*

- Tokenize Bird's Eye View (BEV) representation into LLMs and improve VQA accuracy on DriveLM dataset.
- Semantic distillation from visual encoder to BEV encoder to enable open-vocabulary BEV encoder training.

### Shanghai Artificial Intelligence Laboratory

*OpenMMLab | Mentor: Wenwei Zhang | Manager: Kai Chen*

Shanghai, CN

*Jun 2023 – Aug 2023*

- Supervised Fine-Tuning (SFT) of [InternLM](#) language models.
- Integrated external Python executor to LLM to improve mathematical reasoning capability on GSM8K dataset.

## TOP-TIER PUBLICATIONS

### Summary

2 (1 first-author) in top-tier machine learning conferences (NeurIPS, ICLR)

2 (2 first-author) in top-tier computer vision conferences and journals (TPAMI, ICCV)

3 (1 first-author) in top-tier security conferences (NDSS)

- **FlyTrap: Physical Distance-Pulling Attack Towards Camera-based Autonomous Target Tracking Systems**  
**Shaoyuan Xie**, Mohamad Fakih, Junchi Lu, Fayzah Alshammari, Ningfei Wang, Takami Sato, Halima Bouzidi, Mohammad Al Faruque, Qi Alfred Chen  
*Network and Distributed System Security Symposium (NDSS), 2026*
- **Benchmarking and Improving Bird's Eye View Perception Robustness in Autonomous Driving**  
**Shaoyuan Xie**, Lingdong Kong, Wenwei Zhang, Jiawei Ren, Liang Pan, Kai Chen, Ziwei Liu  
*IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2025*
- **Are VLMs Ready for Autonomous Driving? An Empirical Study from Reliability, Data, and Metric Perspectives**  
**Shaoyuan Xie**, Lingdong Kong, Yuhao Dong, Chonghao Sima, Wenwei Zhang, Qi Alfred Chen, Ziwei Liu, Liang Pan  
*International Conference on Computer Vision (ICCV), 2025 (acceptance rate 24% = 2699/11239)*
- **Can We Trust Embodied Agents? Exploring Backdoor Attacks against Embodied LLM-Based Decision-Making Systems** (\*co-first author)  
Ruochen Jiao\*, **Shaoyuan Xie**\*, Justin Yue, Takami Sato, Lixu Wang, Yixuan Wang, Qi Alfred Chen, Qi Zhu  
*International Conference on Learning Representations (ICLR), 2025 (acceptance rate 32.1% = 3689/11500)*
- **The Heat is On: Understanding and Mitigating Vulnerabilities of Thermal Image Perception in Autonomous Systems**  
Hrushikesh Varma, **Shaoyuan Xie**, Michael Clifford, Qi Alfred Chen, Takeshi Sugawara, Sara Rampazzi  
*Network and Distributed System Security Symposium (NDSS), 2026*
- **Revisiting Physical-World Adversarial Attack on Traffic Sign Recognition: A Commercial Systems Perspective**  
Ningfei Wang, **Shaoyuan Xie**, Takami Sato, Yunpeng Luo, Kaidi Xu, Qi Alfred Chen  
*Network and Distributed System Security Symposium (NDSS), 2025 (acceptance rate 15.4% = 211/1369)*

- **RoboDepth: Robust Out-of-Distribution Depth Estimation under Corruptions**

Lingdong Kong, **Shaoyuan Xie**, Hanjiang Hu, Lai Xing Ng, Benoit R. Cottureau, Wei Tsang Ooi

*Conference on Neural Information Processing Systems (NeurIPS), 2023 (acceptance rate 26.1% = 3218/12343)*

## OTHER PEER-REVIEWED PUBLICATIONS

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- **"Prompter Says": A Linguistic Approach to Understanding and Detecting Jailbreak Attacks Against Large-Language Models**

Dylan Lee, **Shaoyuan Xie**, Shagoto Rahman, Kenneth Pat, David Lee, Qi Alfred Chen

*Proceedings of the 1st ACM Workshop on Large AI Systems and Models with Privacy and Safety Analysis, 2024*

- **On the Adversarial Robustness of Camera-based 3D Object Detection**

**Shaoyuan Xie**, Zichao Li, Zeyu Wang, Cihang Xie

*Transactions on Machine Learning Research (TMLR), 2024*

## RESEARCH IMPACT

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- Email acknowledgements from DJI on autonomous tracking vulnerabilities
- Email acknowledgements from automotive manufacturer on traffic sign recognition vulnerabilities (One company among Tesla, Toyota, Nissan, Mazda, and Hyundai; no direct revelation of the company identity due to the anonymity agreement achieved during the responsible vulnerability disclosure process)

## AWARDS AND HONORS

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- Best Demo, 3rd USENIX Symposium on Vehicle Security and Privacy (2025, *selected for best demo rate 18.2% = 2/11*)
- Honours Bachelor's Degree (2023, *top 1% in univ.*)
- Outstanding Undergraduate Student (2021, *top 1% in univ.*)
- National Scholarship, Ministry of Education of China (2021, *top 0.2% nation-wide*)
- National Scholarship, Ministry of Education of China (2020, *top 0.2% nation-wide*)
- 1st prize in Mathematics Competition of Chinese College Students, Chinese Mathematical Society (2020)
- Provincial Merit Student, Sichuan Provincial Department of Education, (2019, *top 0.4% in high school*)

## ACADEMIC SERVICES

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- *Competition Organizer*: IROS RoboSense (2025), ICRA RoboDrive (2024), ICRA RoboDepth (2023)
- *PC member*: ACM CCS Artifact Evaluation (2024, 2025), NDSS Artifact Evaluation (2025)
- *Reviewer*: CVPR (2023, 2024, 2025), ICCV (2025), TMLR
- *External Reviewer*: USENIX Security (2024, 2025), IEEE S&P (2024), ACM CCS (2024)